



Data Types; Chart Types;

DSCI 454
Week 2, Lecture 1

What are we learning this lecture?



- Data Types and Visual Encoding
 - Categorical
 - Ordinal
 - Time series
 - Numerical
 - Geospatial
- How data type determines chart selection.



Why is data type important?

- Later in the semester, when we perform data analysis, some (no, low, pro) code numerical methods will only work on data of certain types
 - e.g. How do you (or realistically, a computer (working with matrices of numbers)) take the average of: “Monday”, “Wednesday”, “Friday”, or S/M/L?
 - Words/Characters/Strings -> Numbers
- In the next lecture we will also see how using the wrong chart type (intentionally or not) can lead to misunderstanding at best and bias and unethical aspects at worst.

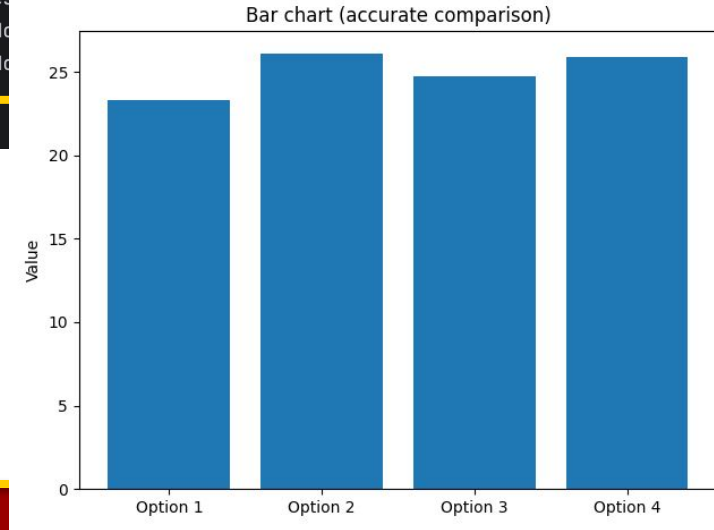
Categorical Data



- Categories
- Labels only
- No order
- e.g. data: smoker, sex, day, meal
- Best Chart Type: Bar
 - Easy to compare options visually at a glance
- Avoid line plots which could imply continuity, scatter OK
- Avoid pie charts which could obfuscate accuracy/precision

	total_bill	tip	sex	smoker	day	time	size	sex_cat	smoker_cat	day_cat	time_cat
0	16.99	1.01	Female	No	Sun	Dinner	2	1	1	3	1
1	10.34	1.66	Male	No	Sun	Dinner	3	0	1	3	1
2	21.01	3.50	Male	No	Sun	Dinner	3	0	1	3	1
3	23.68	3.31	Male	No	Sun	Dinner	2	0	1	3	1
4	24.59	3.61	Female	No	Sun	Dinner	4	1	1	3	1
..	...	...	...	...	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3	0	1	2	1
240	27.18	2.00	Female	Yes	Sat	Dinner	2	1	0	2	1
241	22.67	2.00	Male	Yes							
242	17.82	1.75	Male	No							
243	18.78	3.00	Female	No							

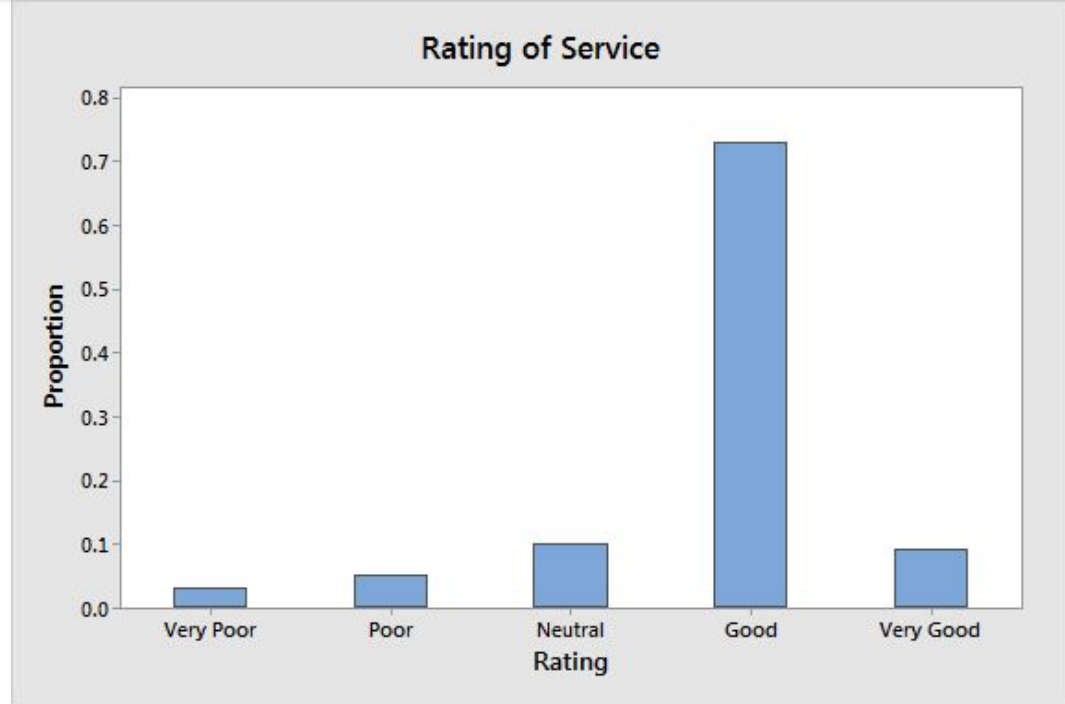
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Ordinal Data



- Ordered
- e.g. data: Survey Results, levels, scales, enums
- Best Chart Type: Ordered Bar
 - Easy to compare options visually at a glance
- Avoid line plots which could imply continuity
- Avoid pie charts which could obfuscate accuracy/precision



<https://statisticsbyjim.com/basics/ordinal-data/>

Knowledge Check



- What's the difference between categorical and ordinal data?
- What type of data is day of the week?
 - Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
 - Thursday, Friday, Saturday, Sunday, Monday, Tuesday
- List 3 categorical data type set examples?
- List 3 ordinal data type set examples?



Time Series Data

- Targets over date/time, line plot where the x axis is date/time
- Best Chart Type: Line
 - Trends
- Order, continuity, spacing all matter
- Can be used w/ aggregations(e.g. rolling avg. etc.) w/ caution

Currency in USD						
Date	Open	High	Low	Close ①	Adj Close ①	Volume
Jan 16, 2026	6,960.54	6,967.30	6,925.09	6,940.01	6,940.01	5,356,550,000
Jan 15, 2026	6,969.46	6,979.34	6,937.93	6,944.47	6,944.47	5,114,050,000
Jan 14, 2026	6,937.41	6,941.20	6,885.74	6,926.60	6,926.60	5,520,820,000
Jan 13, 2026	6,977.41	6,985.74	6,937.93	6,944.47	6,944.47	5,114,050,000

S&P 500

6,940.01

↑ 826.94% +6,191.31 MAX

Jan 16, 4:44:54 PM UTC-5 · INDEXSP · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX

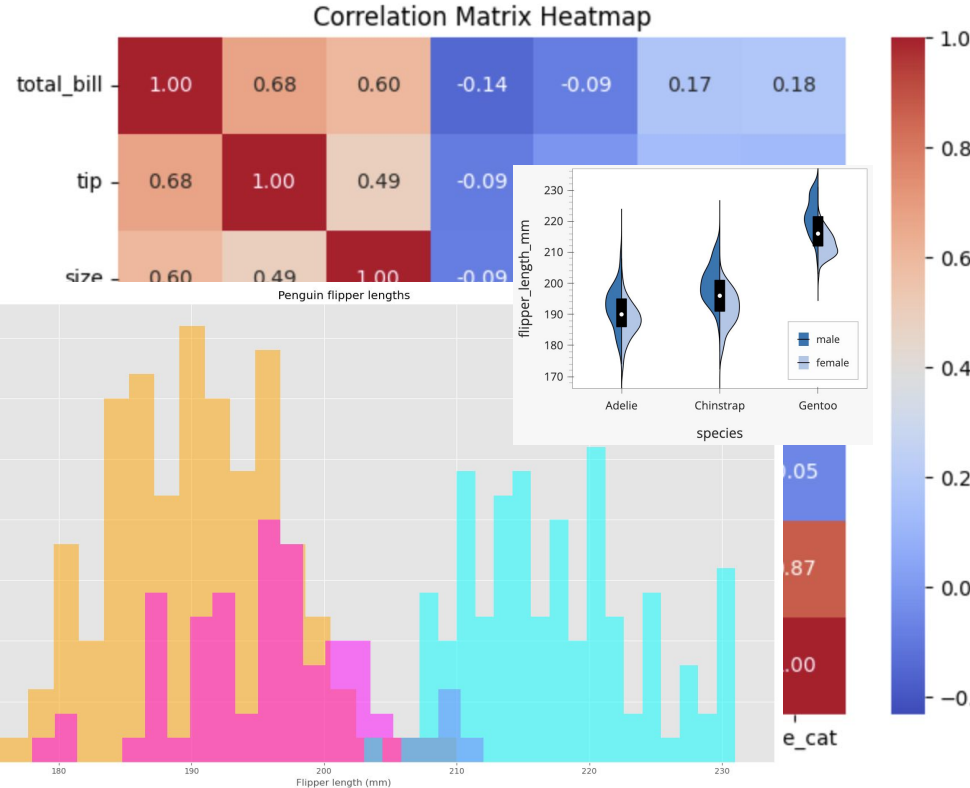
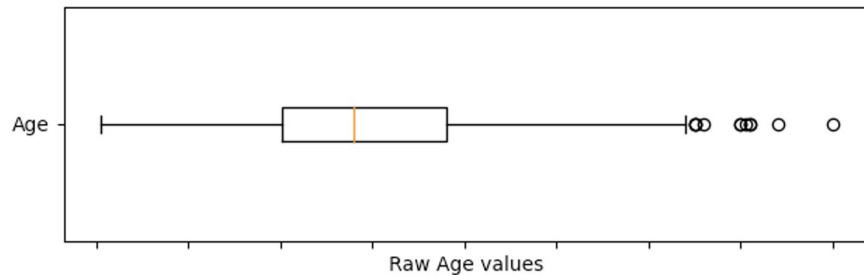


Numerical Data



- Discrete numbers
- Math works (quantitative)
- e.g. data: sales as currency
- Best Chart Type: Histogram, Violin, Heatmaps, Box plot
- Distribution, frequency, outlier detection

Titanic dataset: identifying and removing Age outliers

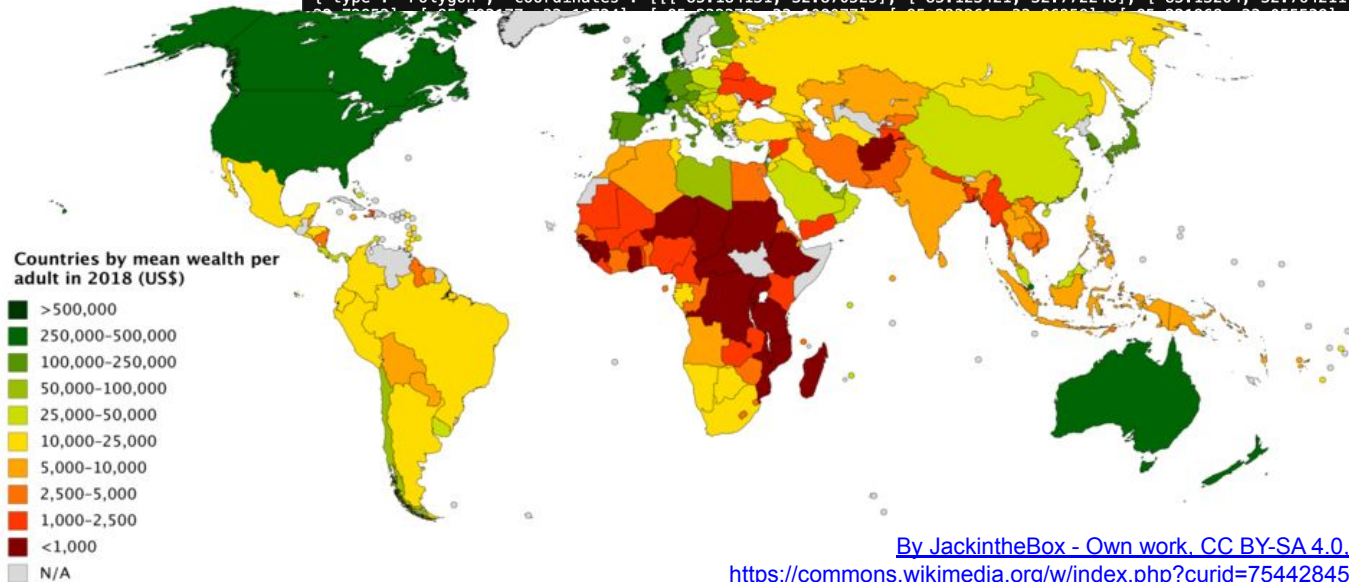


Geospatial Data



	A	B	C	D	E
1	city	city_ascii	lat	lng	country
2	Tokyo	Tokyo	35.687	139.7495	Japan
3	Jakarta	Jakarta	-6.175	106.8275	Indonesia
4	Delhi	Delhi	28.61	77.23	India
5	Guangzhou				
6	Mumbai				
7	Manila				
8	Shanghai				

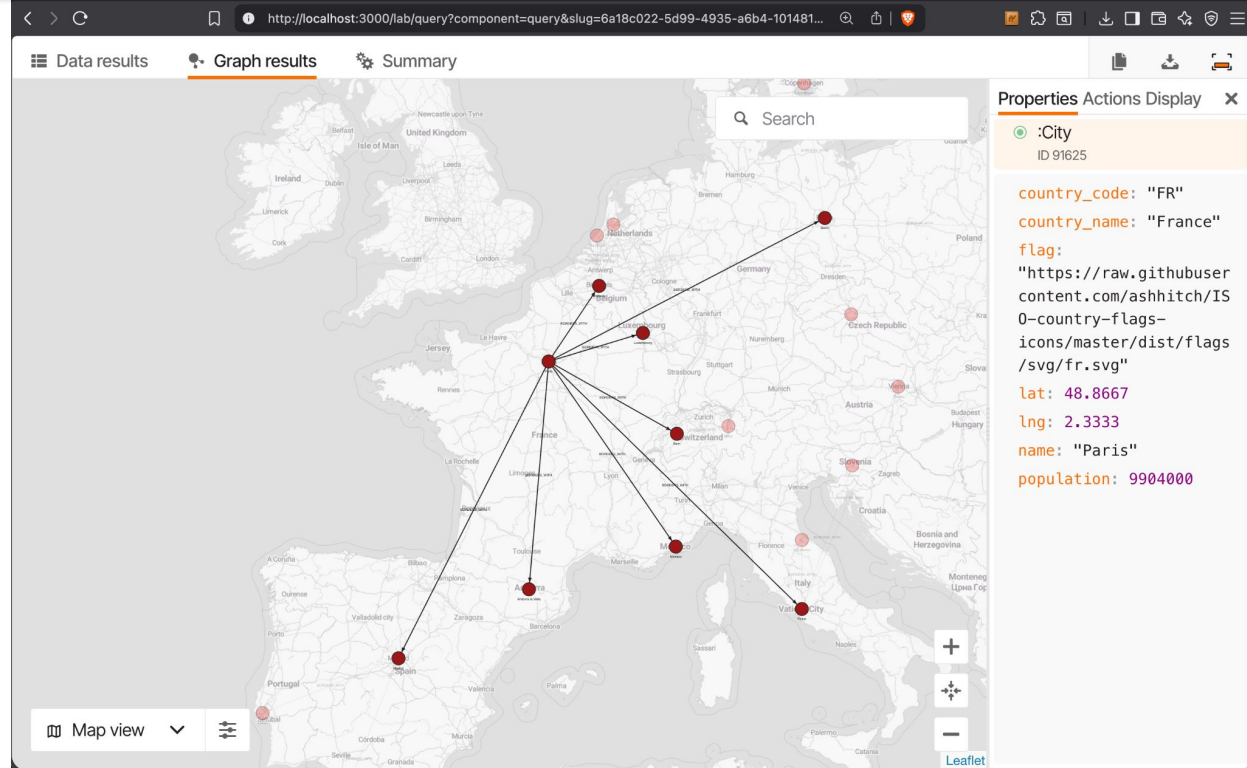
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< > ↻ https://raw.githubusercontent.com/plotly/datasets/master/geojson-counties-fips.json
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Geospatial Example: Memgraph



- Map scatter with links/relationships between points/pins/nodes aka Graph
- More later in the semester



In-Class Exercise



- Data type identification and chart type selection of real-world datasets
 - <https://www.kaggle.com/datasets>
 - <https://datasetsearch.research.google.com/>
 - <https://archive.ics.uci.edu/datasets>
 - <https://scikit-learn.org/stable/api/sklearn.datasets.html>
 - <https://www.kaggle.com/datasets/abdoomoh/all-seaborn-built-in-datasets>
- <https://www.tableau.com/learn/articles/free-public-data-sets>
- <https://neo4j.com/docs/getting-started/appendix/example-data/>
- Memgraph built-in datasets demo

Questions?

